MEET PATEL

+1 (201) 359-1312 | linkedin/meetpatel | contact.mtpatel@gmail.com | Portfolio | Jersey City (Willing to relocate)

EDUCATION

Fairleigh Dickinson University, New Jersey — May 2026
Master's in Computer Science

SKILLS

- **Languages:** Python, SQL, JavaScript
- **ML Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, CUDA, CoreML
- **AI & Architectures:** Transformers, RAG, Multi-Agent Systems (A2A), Model Context Protocol (MCP)
- **MLOps, Tools & Cloud:** Google Cloud, Docker, Kubernetes, CI/CD (Jenkins), Git
- **Certifications & Profiles:** 7+ Courses from HarvardX, Google, Anthropic, Microsoft (Focus: AI, Data Science)

EXPERIENCE

Incoming Deep Learning Intern — May 2026 to July 2026
NJ Department of Health (DOH)

- Selected to **develop and deploy a neural network** to automate the quality classification of newborn screening (NBS) blood spot specimens, identifying specific defect categories.
- Tasked with conducting statistical data analysis to evaluate experimental error rates, **determining the viability of utilizing unsatisfactory biological samples** for diagnostic testing.
- Performing comparative data analysis to identify statistically significant deviations between satisfactory and defective samples, informing future medical testing protocols.

Research Assistant — January 2025 to Present
Fairleigh Dickinson University, New Jersey

- Led research to **resolve complex spatial dependencies** (pixel to pixel relationships) in computer vision models. Performed research on Graph Neural Networks (GNN) and its hybrid versions with CNN, Transformers and Vision Transformer (ViT). Resulting in a paper acceptance at **IEEE** ISEC'26 conference.
- Processed **10,000+ patients' medical visual records** and benchmarked **3 SOTA models** (EfficientNetV2, Vision Transformers and VGG16), developed a high precision pipeline using VGG16, improved accuracy **from 81% to 97.75%** results submitted to **Elsevier Pattern Recognition**.

Machine Learning Intern — January 2023 to April 2023
Tri State Technologies, Ahmedabad, India

- Streamlined **data preprocessing pipelines**, improved training data quality, and reduced model noise.
- Supported the team by developing **ML APIs using FastAPI** and **Jenkins** for CI/CD to connect model inference with frontend interface, ensuring **scalable deployment**.

PROJECTS

Privacy Preserving Yoga Recommendation Model — PyTorch, GNN, on-device computation

- Architected a **privacy centric** AI pipeline that prescribes personalized yoga routines based on real time HRV, Motion and biometric data. Utilized 100% on-device computation to ensure data isolation.
- Engineered a Knowledge Graph containing **190+ yoga poses** from **50+ unstructured textbooks (PDFs)**, resolving semantic conflicts.
- Optimized a GNN for mobile deployment by implementing a knowledge distillation pipeline, generating a lightweight **Core ML** model optimized for the **Apple Neural Engine (ANE)**.

Real-Time Multimodal Q&A System — PyTorch, VLM, ONNX Runtime

- Built a live multimodal question answering system using NLP achieving <**100ms TTFT** by optimizing a **LLaVA style pipeline** with Qwen2-0.5B-Instruct and Apple's FastVLM.
- Optimized the model using **INT8 quantization**, **NPU acceleration** (CoreML) and a ONNX pipeline, achieving a total **4x latency reduction**.

Neuromorphic vs. Transformer NLP Benchmarking — PyTorch, snnTorch, Transformers (Hugging Face)

- Conducted accurate performance benchmarking for **NLP** while comparing a pre-trained **DistilBERT** and a custom-built, from-scratch Spiking Neural Network (SNN) on a 20,000-sample classification task.
- Fine-Tuned SNN for balancing the accuracy and computation power trade-off, the SNN achieved **81% of BERT's accuracy** while being **48x more computationally efficient**.